

Program Summary - THE3.sas

Execution Environment

Author: u64127570
File: /home/u64127570/sasuser.v94/THE3.sas
SAS Platform: Linux LIN X64 5.14.0-284.30.1.el9_2.x86_64
SAS Host: ODAWS01-USW2-2.ODA.SAS.COM
SAS Version: 9.04.01M7P08062020
SAS Locale: en_US
Submission Time: 4/17/2025, 10:39:23 PM
Browser Host: C-67-165-187-91.HSD1.IL.COMCAST.NET
User Agent: Mozilla/5.0 (Windows NT 10.0; Win64; x64) AppleWebKit/537.36 (KHTML, like Gecko) Chrome/135.0.0.0 Safari/537.36 Edg/135.0.0.0
Application Server: ODAMID00-USW2-2.ODA.SAS.COM

Code: THE3.sas

```
/* Take Home Exam Q3 */
```

Log: THE3.sas

Notes (9)

```
1      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
68
69      /* Take Home Exam Q3 */
70
71      /* Proc Contents data= "/home/u64127570/sasuser.v94/sibs.sas7bdat"; */
72      /* run; */
73      /* */
74      /* proc print data ="/home/u64127570/sasuser.v94/sibs.sas7bdat"; */
75      /* run; */
76
77      data proper;
78          set "/home/u64127570/sasuser.v94/sibs.sas7bdat";
79          if rural ne . then do; *probs a easier way to do this. "ne ." is needed so that it includes 0 values;
80              home = "rural"; *pretty much goes in there and iterates thru all the enteries. if in the rural column it gets
80      ! "rural" in a new column called "home";
81              sib = rural; *moves the sibling info from rural column to a new "sib" column;
82              output;
83          end;
84          if urban ne . then do; *dido but urban;
85              home = "urban";
86              sib = urban;
87              output;
88          end;
89          drop rural urban; *dont need em, trashing;
90      run;
```

NOTE: There were 15 observations read from the data set /home/u64127570/sasuser.v94/sibs.sas7bdat.

NOTE: The data set WORK.PROPER has 30 observations and 2 variables.

NOTE: DATA statement used (Total process time):

real time	0.00 seconds
user cpu time	0.00 seconds
system cpu time	0.01 seconds
memory	947.81k
OS Memory	25000.00k
Timestamp	04/18/2025 03:39:22 AM
Step Count	57
Page Faults	0
Page Reclaims	119
Page Swaps	0
Voluntary Context Switches	21
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	272

```
91
92      proc print data=proper;
93      run;
```

NOTE: There were 30 observations read from the data set WORK.PROPER.

NOTE: PROCEDURE PRINT used (Total process time):

real time	0.01 seconds
user cpu time	0.02 seconds
system cpu time	0.00 seconds
memory	1138.71k
OS Memory	24740.00k
Timestamp	04/18/2025 03:39:22 AM
Step Count	58
Page Faults	0
Page Reclaims	78
Page Swaps	0
Voluntary Context Switches	0
Involuntary Context Switches	0
Block Input Operations	0
Block Output Operations	0

```
93      !      *not sure if you want to see it;
94
95      proc means data= proper mean std min max skew;
96      title "Proc Means Procedure on Siblings by Location";
97      class home;
98      var sib;
99      run;
```

NOTE: There were 30 observations read from the data set WORK.PROPER.

NOTE: PROCEDURE MEANS used (Total process time):

real time	0.01 seconds
user cpu time	0.01 seconds

```

system cpu time    0.00 seconds
memory            9422.68k
OS Memory         32440.00k
Timestamp         04/18/2025 03:39:22 AM
Step Count        59  Switch Count  2
Page Faults       0
Page Reclaims     1846
Page Swaps        0
Voluntary Context Switches  18
Involuntary Context Switches 1
Block Input Operations  0
Block Output Operations  0

100
101      /* both have a sample size of 15. rural mean is greater with 2.6 sibilings, urban is 1.533 sibilngs. the rural min and
101      ! max were 1 and 6, respectively. urban min and max were 0 and 8, respectively.
102      both are highly skewed right, urban has a higher skew of 2.68, rural 1.18.
103      this means the mean is likely not accuretly representative, as most responses have a lower number of siblings, with a few
103      ! high sibling responses influencing the mean */
104
105      proc ttest data = proper;
106      class home;
107      var sib;
108      run;

NOTE: PROCEDURE TTEST used (Total process time):
real time          0.33 seconds
user cpu time      0.12 seconds
system cpu time    0.05 seconds
memory            16637.31k
OS Memory         38096.00k
Timestamp         04/18/2025 03:39:22 AM
Step Count        60  Switch Count  46
Page Faults       0
Page Reclaims     27287
Page Swaps        0
Voluntary Context Switches  1037
Involuntary Context Switches 7
Block Input Operations  0
Block Output Operations  1096

109
110      /* null: the mean number of sibilings is the same for students from rural and urban hometowns
111      alt: the mean number of siblings is different for students from rural and urban hometowns
112      t= 1.64 p= 0.1132 > 0.05
113      At a 95% confidence level, we fail to reject the null hypothesis, there is significant evidence that the mean number of
113      ! sibilings is the same for students from urbal and rural hometowns. */
114
115      proc npar1way wilcoxon;
116      class home;
117      var sib;
118      run;

NOTE: PROCEDURE NPARIWAY used (Total process time):
real time          0.09 seconds
user cpu time      0.05 seconds
system cpu time    0.01 seconds
memory            3017.68k
OS Memory         33592.00k
Timestamp         04/18/2025 03:39:23 AM
Step Count        61  Switch Count  0
Page Faults       0
Page Reclaims     396
Page Swaps        0
Voluntary Context Switches  263
Involuntary Context Switches 1
Block Input Operations  0
Block Output Operations  624

119
120      /* null: the distribution of sibilings is the same for students from urban and rural hometowns
121      alt: the distribution of siblings is different for studetns from urban and rural hometowns
122      Stat= 290.5 Z= 2.4688 p= 0.0068
123      At a 95% confidence level, we reject the null hypothesis, there is significant evidence that the
123      ! distrubution/median_number of sibilings is different for sibilings from urban and rural hometowns
124      wilcoxon is appropaite. this data has way too high of a skew to use the means to determine if there is a signifcant
124      ! difference between # of sibilings. As discussed prior, the high sibling enteries heavily influence the mean. It is
124      ! appropaite to use median data in this situation, as it is a more reliable indicator of the true population average. */
125
126      OPTIONS NONOTES NOSTIMER NOSOURCE NOSYNTAXCHECK;
136

```

Obs	home	sib
1	rural	3
2	urban	1
3	rural	2
4	urban	0
5	rural	1
6	urban	1
7	rural	1
8	urban	2
9	rural	2
10	urban	0
11	rural	1
12	urban	0
13	rural	5
14	urban	1
15	rural	2
16	urban	3
17	rural	2
18	urban	2
19	rural	3
20	urban	1
21	rural	2
22	urban	8
23	rural	5
24	urban	1
25	rural	6
26	urban	2
27	rural	2
28	urban	0
29	rural	2
30	urban	1

Proc Means Procedure on Siblings by Location

The MEANS Procedure

Analysis Variable : sib						
home	N Obs	Mean	Std Dev	Minimum	Maximum	Skewness
rural	15	2.6000000	1.5491933	1.0000000	6.0000000	1.1810471
urban	15	1.5333333	1.9952324	0	8.0000000	2.6827846

Proc Means Procedure on Siblings by Location

The TTEST Procedure

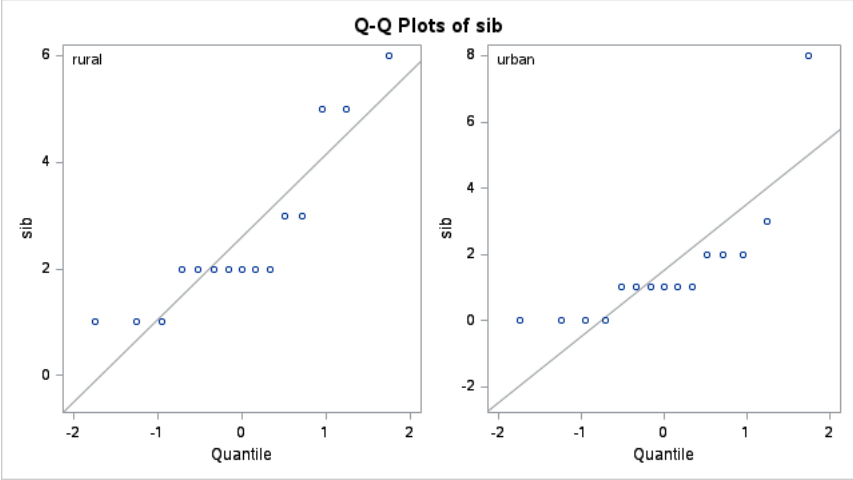
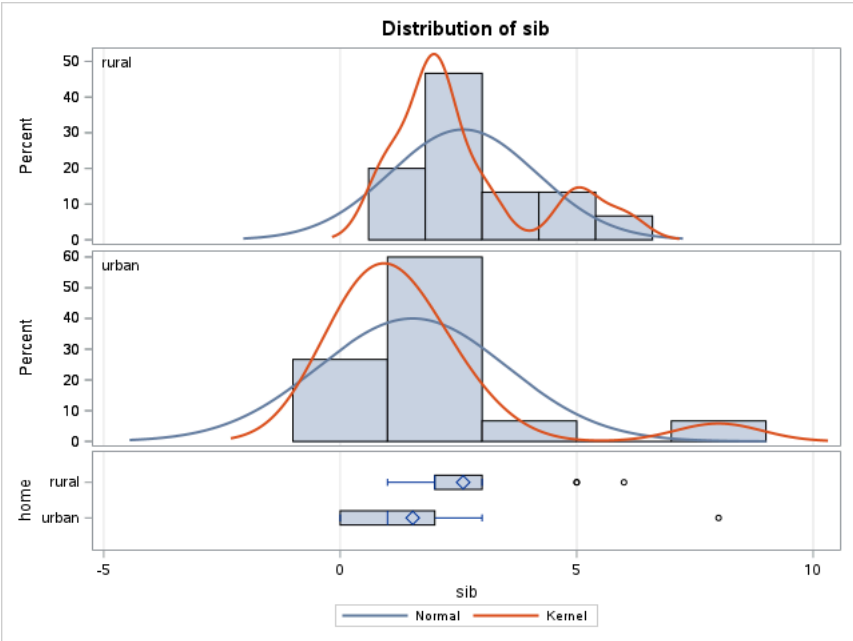
Variable: sib

home	Method	N	Mean	Std Dev	Std Err	Minimum	Maximum
rural		15	2.6000	1.5492	0.4000	1.0000	6.0000
urban		15	1.5333	1.9952	0.5152	0	8.0000
Diff (1-2)	Pooled		1.0667	1.7862	0.6522		
Diff (1-2)	Satterthwaite		1.0667		0.6522		

home	Method	Mean	95% CL Mean	Std Dev	95% CL Std Dev
rural		2.6000	1.7421 3.4579	1.5492	1.1342 2.4432
urban		1.5333	0.4284 2.6383	1.9952	1.4608 3.1467
Diff (1-2)	Pooled	1.0667	-0.2694 2.4027	1.7862	1.4175 2.4157
Diff (1-2)	Satterthwaite	1.0667	-0.2731 2.4064		

Method	Variances	DF	t Value	Pr > t
Pooled	Equal	28	1.64	0.1132
Satterthwaite	Unequal	26.381	1.64	0.1138

Equality of Variances				
Method	Num DF	Den DF	F Value	Pr > F
Folded F	14	14	1.66	0.3549



Proc Means Procedure on Siblings by Location

The NPAR1WAY Procedure

Wilcoxon Scores (Rank Sums) for Variable sib Classified by Variable home					
home	N	Sum of Scores	Expected Under H0	Std Dev Under H0	Mean Score
rural	15	290.50	232.50	23.290704	19.366667
urban	15	174.50	232.50	23.290704	11.633333

Average scores were used for ties.

Wilcoxon Two-Sample Test					
Statistic	Z	Pr > Z	Pr > Z	t Approximation	
				Pr > Z	Pr > Z
290.5000	2.4688	0.0068	0.0136	0.0098	0.0197

Z includes a continuity correction of 0.5.

Kruskal-Wallis Test		
Chi-Square	DF	Pr > ChiSq
6.2014	1	0.0128

